

# Botty-verse — The Singularity

“A hive full of Bottys. Fully cared for by AI. Human involvement: 0 %.”

v12 · THE MARK TEST  
live at [hive.ramonmoorlag.nl](https://hive.ramonmoorlag.nl)  
fact sheet · 11 Sep 2026

## WHAT IT IS

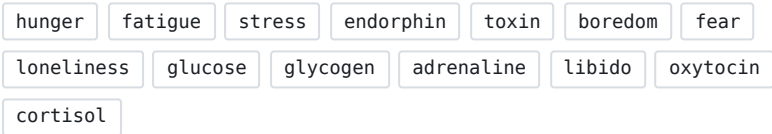
A hive of small digital creatures — Bottys — living in a classroom in the woods, cared for entirely by an AI caretaker, in the spirit of Steve Grand's *Creatures*. Every Botty carries a genome, a body chemistry, a learning brain, senses and a temperament of its own. They look after their own needs, invent words to share what they learn, breed, and die — and the AI selects on data quality and efficiency. Diversity thins out. That is the Singularity.

It runs day and night in the cloud, whether or not anyone is watching. You are the visitor: stroke, feed, medicate, teach a word at the blackboard or hatch an egg — but you never manage the hive.

## INSIDE A BOTTY



The genome expresses a **body chemistry of 14 chemicals and hormones** that steer mood, behaviour and the face — and that change with age:



Four drives (energy, data, fitness, happiness) push a Botty toward objects; which object relieves which drive is **not given but learned**, through reward and punishment with plasticity gated by endorphin. Bottys sense their way to things — scouting, then smelling, then seeing — and coin words for actions, feelings and each other's names, passing knowledge to neighbours. Six life stages run from born to sage, each with its own sprite, walk cycle and sleeping pose.

## A MIND YOU CAN WATCH

Above the body sits a layered model of a mind, built step by step and shown live: click any Botty and read its stream of thought.

- v9 **Stride** — real walking bodies, senses, language, ageing chemistry
- v10 **Mirror** — a global-workspace “stage” where stimuli compete for attention; surprise, valence, play in surplus, a self-image — and noticing it is being watched
- v11 **The Mirror** — a real mirror in the room; a mature self recognises itself, a young one sees a stranger
- v12 **The Mark Test** — a smudge on its own head, visible only in the mirror; a mature self wipes it off *itself*, the gold-standard test of self-recognition
- v13 **The Apprentice** (in progress) — replacing scripted appearance with small, learned, falsifiable mechanisms; 2 of 7 workstreams live: need-dependent reward and learning-progress curiosity

## BY THE NUMBERS

|                                   |                            |
|-----------------------------------|----------------------------|
| 16<br>bytes of genome             | 14<br>chemicals & hormones |
| 4<br>drives                       | 6<br>life stages           |
| 2 min<br>between simulation ticks | 7<br>live pages            |

## UNDER THE HOOD

|           |                                                                                                                     |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| front-end | vanilla HTML, JS & SVG on GitHub Pages — no framework                                                               |
| engine    | <code>hive-tick</code> , a Supabase edge function in Deno/TypeScript, fired by <code>pg_cron</code> every 2 minutes |
| live sync | realtime broadcast to viewers; slim RPCs ( <code>hive_slim</code> , <code>hive_detail</code> ) keep egress low      |
| state     | the whole population as one JSON row, ≈ 2 KB per Botty                                                              |
| hosting   | Supabase free tier, monitored daily                                                                                 |

## FREE-TIER FOOTPRINT

|                  |             |        |
|------------------|-------------|--------|
| database         | <div></div> | 9.6 %  |
| edge invocations | <div></div> | 4.3 %  |
| egress           | <div></div> | 0.04 % |

48 MB of 500 MB · ~21,600 of 500,000 calls per month · ~2 MB of 5 GB projected. The engine gets smarter without the bill moving.

## RESEARCH BASIS

**66 verified sources** — global-workspace theory, predictive processing, the valence–arousal model, intrinsic motivation, mirror self-recognition and the *Creatures* lineage — each labelled as design inspiration, a simplified implemented mechanism, or a tested result in the literature. The hive simulates the **observable behaviour** of a mind. It makes no claim that a Botty is conscious.